Definition : Let (Y, X) be a Random Vector. The Conditional Expectation of Y given $X = x$ is called Regression of Y on X

$$\hat{y} = g(x, \beta) = E(Y \mid X = x)$$

Always Stochastic or Random variable

Can be line, curve, surface, plane, etc.

Can be Stochastic or Non-Stochastic

$(k \times 1)$:

Linear & Non Linear Regression

Consider a data set $D = \{(x_i, y_i) \mid x_i \in \mathbb{R}^k, y_i \in \mathbb{R}, \forall i = 1, 2, \cdots n\}$ where all the x_i's are non-stochastic, but y_i are stochastic & Realised values of Random Variable Y_i's respectively.

Definition :- If the relation, $g(x, \beta)$, between the response variable y & the regressor variable x is Linear in Parameter (β), then it is called Linear Regression.

Eg:- $y = \beta_0 + \beta_1 x + \varepsilon$

$y = \beta_0 + \beta_1 e^{x} + \varepsilon$

Non-Linear Regression - Eg:- $y = \dfrac{1}{\beta_0 + \beta_1 x} + \varepsilon$; $y = \beta_0 \cos(\beta_1 + \beta_2 x) + \varepsilon$

Here, ε is Random error

Varieties of Regression : I.) Linear Model (LM)

1.) Simple Linear Regression
2.) Multiple Linear Regression
3.) Polynomial Regression

II.) Generalised Linear Model (GLM)

1.) Logit / Logistic Model
2.) Probit Model
3.) Poisson - Regression

Others :- Isotonic Regression, Spline Regression etc.

Simple Linear Regression

- Consider a dataset $D = \{(x_i, y_i) \mid x_i \in \mathbb{R}^k, y_i \in \mathbb{R}, \forall i = 1, 2, \cdots n\}$
- x_i's are non stochastic
- y_i are Stochastic & Realised Values of Random Variable Y_i's

Problem Statement: We are interested to have a Prediction line $\hat{y} = g(x, \beta_0, \beta_1) = \beta_0 + \beta_1 x$ which will approximate well the y values if the x values are known.

Minimum distance between the Predicted ($\hat{y}_i$'s) and the true (y_i's) values of Y.

We will consider either Absolute or Square / Euclidean distance.
We will consider values of β_0 & β_1 that will minimise the Distance between the Predicted ($\hat{y}_i$'s) & the true (y_i's) values of Y.

Least Squared Condition: The least squared condition to estimate the model parameters is to minimize $\longrightarrow$ $S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$ with respect to β_0 & β_1

If $(\hat{\beta}_0, \hat{\beta}_1)$ minimizes $S(\beta_0, \beta_1)$, then their values can be obtained by solving the Normal Equations :- $\frac{\partial S}{\partial \beta_0} = 0$ ① & $\frac{\partial S}{\partial \beta_1} = 0$ ②

Solving ① $\rightarrow -2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \Rightarrow \sum y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i$

Solving ② $\rightarrow -2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0 \Rightarrow \sum x_i y_i = \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2$

Least Squares Normal Eqn's

On solving these we find :- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, Here :- $\bar{y} = \frac{1}{n}\sum y_i$ & $\bar{x} = \frac{1}{n}\sum x_i$

& $\beta_1 = \dfrac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = \dfrac{\sum y_i (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \dfrac{S_{xy}}{S_{xx}}$

$\therefore$ The Prediction line will be :- $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

The difference between the observed value y_i & the corresponding fitted value $\hat{y}_i$ is a Residual.

$$e_i = y_i - \hat{y}_i$$

Vector Notation

- Construct a data set $D = \{(x_i, y_i) \mid x_i \in \mathbb{R}, y_i \in \mathbb{R}\}$
- x_i's are Non Stochastic
- y_i's are Stochastic

$y = (y_1, y_2, \cdots y_n)'$, $x = (x_1, x_2, \cdots x_n)'$, $\beta = (\beta_0, \beta_1)'$

$\mathbb{1} = (1, 1 \cdots 1)^T$

Problem Statement: We are interested to have a prediction vector:

$\hat{y} = \underbrace{\boxed{g(x, \beta)}}_{}$ = $[\mathbb{1} \; x] \; \beta$ g.L.M.

It becomes a problem in $\mathbb{R}^n$ now

$y = f(x) + \varepsilon$ $\underset{\sim}{Y} =$, $\underset{\sim}{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$, $X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} = [\underset{\sim}{1} \; \underset{\sim}{x}]$

$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$

$$\therefore \quad \boxed{\underset{\sim}{Y} = X \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}}$$

If $\hat{y} = g(x_1, x_2, \beta)$ Then :-

$\underset{\sim}{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$, $\underset{\sim}{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$, $\underset{\sim}{\varepsilon}$ same as before

& $X = \begin{pmatrix} 1 & x_{11} & x_{21} \\ 1 & x_{12} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{1n} & x_{2n} \end{pmatrix} = [\underset{\sim}{1} \; \underset{\sim}{x}_1 \; \underset{\sim}{x}_2]$

So,

$$\boxed{\underset{\sim}{Y} = X \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}}$$

or $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$

This is Multiple Linear Regression.

Polynomial Regression

$\underset{\sim}{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_5 \end{pmatrix}$; $\underset{\sim}{X} = [\underset{\sim}{1} \; \underset{\sim}{x}_1 \; \underset{\sim}{x}_2 \; \underset{\sim}{x_1 x_2} \; \underset{\sim}{x}_1^2 \; \underset{\sim}{x}_2^2] = \begin{pmatrix} 1 & x_{11} & \cdots \\ 1 & x_{12} & \cdots \\ 1 & \vdots & \\ \vdots & \vdots & \\ 1 & \cdots & \cdots \end{pmatrix}$

It is also a Linear Regression

$$\underset{\sim}{Y} = X \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$ or $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \beta_4 x_{1i}^2 + \beta_5 x_{2i}^2 + \varepsilon_i$

All of the above are Linear Models.

Vector Space :- $(V, +, \cdot)$

A vector space V over real numbers $\mathbb{R}$ is a collection of vectors such that

1.) $+ : V \times V \rightarrow V$ (Closed under vector addition)
2.) $(x+y) + z = x + (y+z)$ for all $x, y, z \in V$ (Associative)
3.) There exists $0 \in V$ such that :- $0 + x = x + 0 = x$ for all $x \in V$ [Identity element exists]

4.) There exists $-x \in V$ for each x such that $(-x) + x = x + (-x) = 0$ [Inverse Exists]
5.) $x + y = y + x$ [Commutative]
6.) $a \cdot (b \cdot x) = (ab) \cdot x \quad \forall a, b \in \mathbb{R} \ \& \ x \in V$
7.) $1 \cdot x = x \cdot 1 = x \quad \forall x \in V$ [Multiplicative Identity]
8.) $(a+b) \cdot x = (a \cdot x) + (b \cdot x) \quad \forall a, b \in \mathbb{R} \ \& \ x \in V$
9.) $a \cdot (x+y) = a \cdot (x) + a \cdot (y)$

Sub-Space $(S, +, \cdot)$:-

If a subset S of V is a Vector Space itself then S is called Subspace of V.

How to check S is a Subspace of V? 1.) Whether $0 \in S$
2.) Whether $x + a \cdot y \in S \quad \forall x, y \in S \ \& \ a \in \mathbb{R}$

Example :- 1.) All lines passing through $(0,0)$ in $\mathbb{R}^2$
2.) All planes passing through origin in $\mathbb{R}^n$
3.) $\mathbb{P}_5$ in $\mathbb{P}_7$ — P_n is polynomial with degree upto n

$\mathbb{P}_5 \rightarrow (\beta_0, \beta_1, \dots \beta_5, 0, 0)$, $\mathbb{P}_7 \rightarrow (\beta_0, \beta_1, \beta_2 \dots \beta_7)$

Span :-

The Span of a set vectors $\{v_1, v_2, \dots v_k\} \in V$ is the collection :-

$$Sp\{v_1, v_2, \dots v_k\} = \left\{ \sum_{i=1}^{k} c_i v_i \;\middle|\; c_i \in \mathbb{R} \right\}$$

which is the collection of all possible linear combinations of $\{v_1, v_2 \dots v_k\}$

Note :- A span is always a subspace.

Example $\longrightarrow$ a.) $Sp\{(0,1),(1,1)\} = Sp\{(0,1),(1,1)\} = \mathbb{R}^2$
b.) $Sp\{(0,1,0),(1,0,0)\} = \mathbb{R} \times \mathbb{R} \times \{0\} =$ xy plane in $\mathbb{R}^3$

Linear Independence

:- A set of vectors $\{v_1, v_2 \cdots v_n\} \in V$ are said to be linearly independent iff $\sum c_i v_i = 0 \Rightarrow c_1 = c_2 = \cdots c_n = 0$. On the other hand, if $\sum c_i v_i = 0$ holds for some non-zero $c_i \in \mathbb{R}$, the vectors are called Linearly dependent.

Example :- a.) $\{(0,1),(1,0)\}$ are Independent
b.) $\{(0,1),(1,1)\}$ are Independent
c.) $\{(0,1),(1,0),(1,1)\}$ are Dependent.

Basis

:- If $\{v_1, v_2 \cdots v_k\}$ are linearly independent, then it is the basis of $Sp\{v_1, v_2 \cdots v_k\}$, and the dimension of $Sp\{v_1, v_2 \cdots v_k\}$ is the number of linearly independent elements in $\{v_1, v_2 \cdots v_k\}$

Example :- a.) $\{(0,1),(1,1)\}$ is a basis of $\mathbb{R}^2$; Dimension $= 2$
b.) $\{(0,1),(1,0)\}$ is a basis of $\mathbb{R}^2$ also. Dimension $= 2$
c.) $\{(0,1),(1,0),(1,1)\}$ is NOT a basis of $\mathbb{R}^2$.

Orthogonal Vectors

Two vectors $u, v \in V$ are said to be orthogonal if $u^T v = \sum u_i v_i = 0$

Orthogonal Complement :- If $S \subseteq V$ is a subspace then the Orthogonal Complement of S, denoted by $S^\perp$ is a collection :-

$$S^\perp = \{v \mid v \in V,\ u^T v = 0,\ \forall u \in S\}$$

and $\dim(S^\perp) = \dim(V) - \dim(S)$

Example :- a.) $Sp\{(1,0,0,0),(0,0,1,0)\} \perp Sp\{(0,1,0,0),(0,0,0,1)\}$
b.) $Sp\{(1,1,0,0),(0,1,1,0),(1,0,1,0)\} \perp Sp\{(0,0,0,1)\}$

Remarks

:- 1.) Basis is Not unique.
2.) Elements of a basis need not be orthogonal to each other
3.) Linear Independence need not imply orthogonality.
4.) Orthogonality implies independence.
5.) Orthogonal vectors with unit length are called Orthonormal Vectors

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Projection

1.) If $S \subseteq V$, then the Projection matrix of subspace S is P_s satisfying :-
a.) $P_s v = v$ if $v \in S$

b.) $P_s v \in S$ for all $v \in V$

2.) A Projection Matrix P_s is an Orthogonal Projection Matrix of subspace $S \subseteq V$ if $(I - P_s)$ is a projection matrix of $S^{\perp} \subseteq V$ too.

3.) If $\{v_1, v_2, \dots v_k\}$ is an Orthonormal Basis of the subspace $S \subseteq V$ then the orthogonal projection matrix of S is $P_s = \sum v_i v_i^T$

Idempotency

1.) If a matrix P satisfies the relation that $P^2 = P$, then P is called an Idempotent Matrix.

2.) An Idempotent matrix has Eigenvalues 0 & 1.
3.) A Projection Matrix is an Idempotent Matrix

Note :- In regression, $\hat{y}$ eventually becomes the orthogonal Projection of $y \in \mathbb{R}^n$ in the subspace $S = Sp\{1, x\}$

Column Space

The column space of a matrix $A = [a_1, a_2 \cdots a_n]$ with columns $a_1, a_2 \cdots a_n$ is :-

$$C(A) = Sp\{a_1, a_2, a_3 \cdots a_n\} = \{Ax \mid x \in \mathbb{R}^n\}$$

Hence, Row Space of A denoted by $R(A) = C(A^T)$

Properties :- 1.) $C(A : B) = C(A) + C(B)$
2.) $C(AB) \subseteq C(A)$
3.) $\dim[C(A)] = \text{Rank}(A)$
4.) $C(AA^T) = C(A) \implies \text{Rank}(AA^T) = \text{Rank}(A)$
5.) If A has n-rows then $\dim(C(A)^{\perp}) = n - \text{Rank}(A)$

Quadratic Forms :-

A square symmetric matrix $A = ((A_{ij}))_{n \times n}$ is said to be -

a.) Positive Definite (p.d.) if $x^T A x > 0$ for all $x \neq 0 \in \mathbb{R}^n$
b.) Positive Semi-Definite (p.s.d) if $x^T A x \geq 0$ for all $x \neq 0 \in \mathbb{R}^n$
[Also called non-negative definite (n.n.d.)]

Properties - a.) If A is p.d. then $|A| > 0$
b.) If A is p.s.d. then $|A| \geq 0$

Generalised Inverse :-

A matrix G is said to be a generalize inverse of a matrix A if $AGA = A$. Usually G is denoted by A^{-}

Properties – 1.) If A is $m \times n$ then A^{-1} is $n \times m$
2.) A^{-1} is not unique.
3.) For a matrix A, the Projection Matrix of $C(A)$ is AA^{-1}
4.) For a matrix A, the orthogonal Projection Matrix of $C(A)$ is $A(A^TA)^{-1}A^T$.

Note – In regression, the prediction $\hat{y}$ and the error $y - \hat{y}$ are Orthogonal to each other.

Let $X = (X_1, X_2, \dots X_n)^T$ be a Random Vector with finite expectation for each of the component, then we define Expectation of X as :-

$$E(X) = (E(X_1), E(X_2), \dots E(X_n))^T$$

We know :- $E(X) = \begin{cases} \sum_{i=-\infty}^{\infty} x_i \, p(x_i) & \text{if } \sum_{i=-\infty}^{\infty} |x_i| \, p(x_i) < \infty \\ \int_{-\infty}^{\infty} x \, p(x) \, dx & \text{if } \int_{-\infty}^{\infty} |x| \, p(x) \, dx < \infty \end{cases}$

Similarly, if $Y = ((Y_{ij}))_{m \times n}$ is a random matrix with finite expectation for each of the component, then we define Expectation of a random matrix as :-

$$E(Y) = ((E(Y_{ij})))_{m \times n}$$

$\underset{\sim}{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$ then $E(\underset{\sim}{X}) = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{pmatrix}$ if $E(X_i)$ exists for all $i = 1, \dots n$

Note – Component wise Median is not the median of the random vector

Dispersion Matrix :-

It is also called Variance - Covariance Matrix

$$D(X) = ((Cov(X_i, X_j)))_{n \times n}, \quad Cov(X, X) = E[(X - E(X))(X - E(X))^T]$$

We know $Var(X) = E(X^2) - (E(X))^2 = E(X - E(X))^2$

$Cov(X, Y) = E(XY - E(X)E(Y))$

$= E((X - E(X))(Y - E(Y))) = Cov(Y, X)$

$\therefore Cov(X, X) = Var(X)$

Note :- 1.) $Cov(V_p, V_q) = Cov((V_i, V_j))_{p \times q}$
2.) $E(X + b) = E(X) + b$
3.) $D(X + b) = D(X)$
4.) $Cov(X + b, Y + c) = Cov(X, Y)$

Let $E(\underset{\sim}{X}) = \underset{\sim}{\mu}$; $D(\underset{\sim}{X}) = E[(\underset{\sim}{X} - E(\underset{\sim}{X}))(\underset{\sim}{X} - E(\underset{\sim}{X}))^T]$

$= E[(\underset{\sim}{X} - \underset{\sim}{\mu})(\underset{\sim}{X} - \underset{\sim}{\mu})^T]$

$= ((Cov(X_i, X_j)))_{i \times j}$

$= E(\underset{\sim}{X} \underset{\sim}{X}^T) - \underset{\sim}{\mu} \underset{\sim}{\mu}^T$

Note - 1.) Expectation is a location equivariant operation
2.) Dispersion is a location invariant operation

Let X be a Random vector with n - components such that $E(X) = \mu$ & $D(X) = \Sigma$ then :-

1.) $E(l^T X) = l^T \mu$, where $l \in \mathbb{R}^n$ is a constant vector.

$$E(\underset{\sim}{l}\, \underset{\sim}{X}) = E\left(\sum_{i=0}^{n} l_i X_i\right) = \sum l_i E(X_i) = \sum l_i \mu_i = \underset{\sim}{l}^T \underset{\sim}{\mu}$$

2.) $E(\underset{m\times n}{A}\, \underset{n\times 1}{\underset{\sim}{X}}) = E(\underset{m\times 1}{Y}) = A \underset{\sim}{\mu}$; A is $\mathbb{R}^{p\times n}$ constant matrix

3.) $D(l^T X) = l^T \Sigma l$

4.) $D(AX) = A \Sigma A^T$ & $Cov(AX, BX) = A \Sigma B^T$

5.) $Cov(V_{p_1}, V_q) = I$, then $Cov(AV, BV) = A\, I\, B^T$

Proof :- $D(\underset{\sim}{l}^T \underset{\sim}{X}) = Cov(\underset{\sim}{l}^T \underset{\sim}{x}, \underset{\sim}{l}^T \underset{\sim}{x}) = Cov(\sum l_i x_i, \sum l_j x_j)$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} l_i l_j\, Cov(X_i, X_j) = \sigma_{ij}$$

$$= \underset{\sim}{l}^T \Sigma \underset{\sim}{l}$$

Let $E(X) = \mu$, $D(X) = \Sigma$; It is easy to notice that $\Sigma = \Sigma^T$

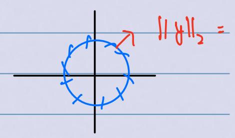
$D(X)$ will be a P.S.D. matrix.

Proof :- $Y = a^T X$; $X \in \mathbb{R}^n$ & $a \neq 0 \in \mathbb{R}^n$

$\Rightarrow Var(Y) \geq 0$; $\therefore a^T \Sigma a \geq 0 \;\; \forall a \neq 0$

So, Σ is a p.s.d.

Let $A = A^T$; $\Rightarrow$ $\boxed{\lambda_{max} = \sup_{\underset{\sim}{x} \neq 0} \frac{\underset{\sim}{x}^T A \underset{\sim}{x}}{\underset{\sim}{x}^T \underset{\sim}{x}} = \sup_{\|y\|_2 \leq 1} \underset{\sim}{y}^T A \underset{\sim}{y}}$ $x \neq 0$



Similarily , $\lambda_{min} = \inf_{\underset{\sim}{x} \neq 0} \frac{\underset{\sim}{x}^T A \underset{\sim}{x}}{\underset{\sim}{x}^T \underset{\sim}{x}} = \inf_{\|y\|_2 \leq 1} \underset{\sim}{y}^T A \underset{\sim}{y}$

Note :- $[0, 1] \longrightarrow$ $0 =$ min & $1 =$ max

$(0, 1] \longrightarrow$ min = Undefined & $1 =$ max

Greatest Lower Bound $\Rightarrow$ Inf

Similarily , Lowest Upper Bound $\Rightarrow$ Sup ; Inf & Sup need not belong to the set.

Theorem - 1

Let X be a random vector with n-components such that $E(X) = \mu$ & $D(X) = \Sigma$. Then $\longrightarrow$

$$E(X^T A X) = tr(\Sigma A) + \mu^T A \mu$$

Applications :- Case 1 Let $A = I_n$ $\therefore$ $E(X^T A X) = E(X^T X)$

$= E(\sum X_i^2) = \sum_{i=1}^{n} E(X_i^2) = \sum (\sigma_i^2 + \mu^2)$

$= \sum \sigma_i^2 + \sum \mu^2 \rightarrow \mu^T I_n \mu$

$tr(\Sigma I_n)$

Case - 2 : $X_i \overset{iid}{\sim} N(0,1)$ $\Rightarrow \Sigma = I_n$; $A = I_n$, $\mu_i = 0 \; \forall i$

It is clear that $\sum_{i=1}^{n} X_i^2 \sim \chi^2_n \rightarrow$ Chi - Squared Distribution

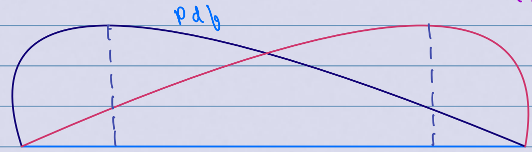
$E(\sum_{i=1}^{n} X_i^2) = n$

Now Consider $E(X) = \mu \neq 0 \Rightarrow E(\sum X_i^2) = n + \mu^T \mu > n$

$X_i \overset{iid}{\sim} N(0,1)$; $Q_1 = X^T X = \sum_{i=1}^{n} X_i^2$. So $E(Q_1) = \sum_{i=1}^{n} X_i^2 = n$

Let $Y_i \overset{iid}{\sim} N(\mu_i, 1)$; $Q_2 = Y^T Y = \sum_{i=1}^{n} Y_i^2$. So $E(Q_2) = \underbrace{n}_{\text{Degrees of Freedom}} + \underbrace{\mu^T \mu}_{\text{Non Central Parameter}}$

p d f



$E(Q_1) < E(Q_2)$ by a factor called Non - Central Parameter

Now, $E(X^T A X) = E(tr(X^T A X)) = E(tr(A X X^T)) = tr(A E(X X^T))$

Real No.

$= tr[A(\Sigma + \mu \mu^T)]$

$= tr(A \Sigma) + \mu^T A \mu$

Theorem - 2

Let X be a random vector with n-components such that $E(X) = \mu$ & $D(X) = \Sigma$. Then $\longrightarrow$

$$P(X - \mu) \in C(\Sigma) = 1$$

In simple terms it tells us that Probability of Deviation / Fluctuation belonging to the Column Space of Variance - Covariance is 1.

Proof :- Let $C(A)$ be all possible linear combinations of the Columns of A matrix.

Also, let $\ell \in [C(\Sigma)]^{\perp} \rightarrow$ Orthogonal Component of $C(\Sigma)$

$\therefore \ell^T \Sigma = 0 \Rightarrow \ell^T \Sigma \ell = 0$

So, $D(\ell^T X) = 0$. Now we know that $D(X + b) = D(X)$

$\therefore D(\ell^T (X - \mu)) = 0 \Rightarrow Var(\ell^T (X - \mu)) = 0$ & $E(\ell^T (X - \mu)) = 0$

This means for $X - \mu$, mean & var are both 0

So, $P(\ell^T (X - \mu) = 0) = 1$ OR $P[\ell \perp (X - \mu)] = 1$

$\therefore$

$$\boxed{P(X - \mu) \in C(\Sigma) = 1}$$

Multivariate Normal

A random vector X is said to follow Multivariate Normal $N(\mu, \Sigma)$ if it has density:-

$$f(x) = \frac{\exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right]}{(\sqrt{2\pi})^n \sqrt{|\Sigma|}}$$

for some $\mu \in \mathbb{R}^n$ & p.s.d. Σ

① If $X \sim N(\mu, \Sigma)$, then $AX \sim N(A\mu, A\Sigma A^T)$

Q. 1.) Let $(X, Y) \sim$ Bivariate Normal $(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho_{xy})$. Then obtain the pdf $f(x,y)$

$\underset{\sim}{Z} = \begin{pmatrix} x \\ y \end{pmatrix}$ $\quad \underset{\sim}{\mu} = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}$; $\Sigma = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_y\sigma_x & \sigma_y^2 \end{pmatrix}$

$$\therefore f(\underset{\sim}{Z}) = \frac{\exp\left[-\frac{1}{2}(\underset{\sim}{Z}-\underset{\sim}{\mu})\, \Sigma^{-1}\, (\underset{\sim}{Z}-\underset{\sim}{\mu})^T\right]}{(\sqrt{2\pi})^2 \sqrt{|\Sigma|}}$$

$\underset{\sim}{Z} - \underset{\sim}{\mu} = \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix}$; $\Sigma^{-1} = \frac{1}{(1-\rho^2)\sigma_x^2\sigma_y^2} \begin{pmatrix} \sigma_y^2 & -\rho\sigma_y\sigma_x \\ -\rho\sigma_x\sigma_y & \sigma_x^2 \end{pmatrix}$

$(\underset{\sim}{Z} - \underset{\sim}{\mu})^T = (x-\mu_x \quad y-\mu_y)$

$$\text{So, } f(x,y) = \frac{\exp\left[-\frac{1}{2}\left(\frac{1}{1-\rho^2}\right)\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)\right]\right]}{2\pi\left(\sqrt{1-\rho^2}\right)\sigma_x\sigma_y}$$

We know that $\chi_n^2 = \text{gamma}\left(\overset{\alpha}{\frac{n}{2}}, \overset{\lambda}{\frac{1}{2}}\right)$; $E(g) = \frac{\alpha}{\lambda} = n$ & $V(g) = \frac{\alpha}{\lambda^2} = 2n$

$$f(x) = \frac{\lambda^\alpha e^{-\lambda x} x^{\alpha-1}}{\sqrt{\alpha}}$$

Theorem

If $X \sim N(\mu, I_n)$, then $X^T A X$ has Chi-Squared Distribution iff A is Idempotent $[A^2 = A]$. Moreover Degrees of Freedom $=$ Rank(A) & Non Central Parameter $= \mu^T A \mu$

This theorem is called Cochran's Theorem

Corollary :-

If A & A_1 are symmetric & idempotent matrices such that $A = A_1 + A_2$ & A_2 is a p.s.d matrix, then $X^T A_1 X$ & $X^T A_2 X$ are independently distributed

Example :- $\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \sim N(0, I_3)$

$A = I_3 = A_1 + A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$X^T A X = X^T (A_1 + A_2) X = X^T A_1 X + X^T A_2 X$

$\Rightarrow \sum_{i=1}^{3} X_i^2 = \sum_{i=1}^{2} X_i^2 + X_3^2$

L.H.S $\to \sum_{i=1}^{3} X_i^2 \sim \chi^2_{df=3,\ ncp=0}$; R.H.S $\to X^T A_1 X \sim \chi^2_{df=2,\ nc=0}$ $X^T A_2 X \sim \chi^2_{df=1,\ ncp=0}$

$\therefore X^T A_1 X$ & $X^T A_2 X$ are Independently - distributed

Theorem

Let $X \sim N(\mu, I_n)$ & A is symmetric & $CA = 0$, then $X^T A X$ & CX are independtly distributed

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T-Statistic

Let $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$. Define :- $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$

& $S^2 = \sum_{i=1}^{n} (X_i - \bar{X})^2$

① Find Distribution of $\bar{X}$ & S^2

$X \sim N(\mu, \sigma^2 I_n)$; $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i^2 = \underbrace{\left(\frac{1}{n}\right) \underset{\sim}{1}^T}_{C} \underset{\sim}{X}$

$S^2 = \sum (X_i - \bar{X})^2 = X^T \underbrace{\left[I_n - \frac{1}{n} 1 1^T \right]}_{A} X$

It is easy to verify that :- ① $S^2 = X^T A X$ ② $A^T = A$, ③ $A^2 = A$ ④ Rank$(A) = n-1$

Now, $CX \sim N(C\mu, C\Sigma C^T)$

$C\mu = \frac{1}{n} \times 1^T \underset{\sim}{\mu} = \frac{1}{n} 1^T (\mu) 1 = \mu$

$C \Sigma C^T = \frac{1}{n^2} 1^T \Sigma 1 = \frac{1}{n^2} 1^T (\sigma^2 I_n) 1 = \frac{\sigma^2}{n}$

Now, $X^T A X = \sigma^2 \left(\frac{X}{\sigma}\right)^T A \left(\frac{X}{\sigma}\right)$

Here, we can see that $\rightarrow \frac{X}{\sigma} \sim N\left(\frac{\mu}{\sigma}, 1 \cdot I_n\right)$ μ^*

$\therefore$ We can apply Cochran's Theorem

$$\sigma^{-2}\left[\left(\frac{X}{\sigma}\right)^T A \left(\frac{X}{\sigma}\right)\right] \sim \sigma^{-2} \chi^2_{d.f. = n-1}$$

$n.c.p = \mu^{*T} A \mu = 0$

② Show that they are i.d. ; $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow \bar{X} - \mu \sim N\left(0, \frac{\sigma^2}{n}\right)$

$\therefore \frac{\sqrt{n}}{\sigma}(\bar{X} - \mu) \sim N(0,1)$

Data :- It is a paired collection of $\{(y_i, x_i) \mid i = 1, 2 \cdots n\}$

Model : ① $y_i = \boxed{\beta_0 + \beta_1 x_i + \varepsilon_i}$ for $i = 1, 2 \cdots n$

Assumption

② $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$ ③ x_i's are Non stochastic

Unknown :- β_0, β_1 & σ^2 have to be estimated

Matrix Representation :- $\underset{\sim}{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$; $\underset{\sim}{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$; $\underset{\sim}{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$

$\underset{n \times 2}{X} = [\underset{\sim}{1} \; \underset{\sim}{x}] = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$; Model : $\underset{\sim}{Y} = X \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$

Here, $\underset{\sim}{\varepsilon} \sim N(0, \sigma^2 I_n)$ & $(\underset{\sim}{\beta}, \sigma^2)$ are to be estimated

In general, $X = [\underset{\sim}{1} \; \underset{\sim}{x}_1 \cdots \underset{\sim}{x}_m]$; $\underset{\sim}{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}$

$\underset{(n\times 1)}{\underset{\sim}{Y}} = \underset{(n\times m)}{X} \underset{(m\times 1)}{\underset{\sim}{\beta}} + \underset{(n\times 1)}{\underset{\sim}{\varepsilon}}$; $(\underset{\sim}{Y} - X\underset{\sim}{\beta}) \sim N(\underset{\sim}{0}, \sigma^2 I_n)$

Least Squared Condition :- $S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2$

$= (\underset{\sim}{Y} - X\underset{\sim}{\beta})^T (\underset{\sim}{Y} - X\underset{\sim}{\beta}) = \underset{\sim}{Y}^T Y - \underset{\sim}{Y}^T X \underset{\sim}{\beta} - \underset{\sim}{\beta}^T X^T \underset{\sim}{Y} + \underset{\sim}{\beta}^T X^T X \underset{\sim}{\beta}$

$S(\underset{\sim}{\beta}) = \underset{\sim}{Y}^T \underset{\sim}{Y} - 2\underset{\sim}{Y}^T X \underset{\sim}{\beta} + \underset{\sim}{\beta}^T (X^T X) \underset{\sim}{\beta}$

$\frac{\partial S}{\partial \underset{\sim}{\beta}} = \underset{\sim}{0} \Rightarrow X^T \underset{\sim}{Y} = (X^T X) \underset{\sim}{\beta} \rightarrow$ Normal Eq^n

If exists

$$\therefore \ \hat{\underset{\sim}{\beta}}_{L.S.} = \boxed{(X^T X)^{-1}} X^T Y$$

Maximum Likelihood Estimation :- $f(y) = \dfrac{e^{\boxed{-\frac{1}{2}(\underset{\sim}{Y} - X\underset{\sim}{\beta})^T \frac{I_n}{\sigma^2}(\underset{\sim}{Y} - X\underset{\sim}{\beta})}}}{(\sqrt{2\pi})^n \sqrt{|\sigma^2 I_n|}}$ → Part we care about

$$\ln(f(y)) = C - \frac{1}{2\sigma^2}\underbrace{\boxed{(\underset{\sim}{Y} - X\underset{\sim}{\beta})^T(\underset{\sim}{Y} - X\underset{\sim}{\beta})}}_{S(\underset{\sim}{\beta})}$$

Under Normality → $\hat{\beta}_{MLE} = \hat{\beta}_{L.S.}$

$$Y = X\beta + \varepsilon \quad ; \quad \varepsilon \sim N(0, \sigma^2 I_n)$$

⇒ **Gauss Markov model** : $\boxed{Y \sim N(X\beta, \sigma^2 I_n)}$ → Holds for all linear models

$$\hat{\beta}_{MLE} = \hat{\beta}_{L.S} = \hat{\beta} = (X^T X)^{-1}(X^T Y).$$

Predicted Vector, $\hat{y}$ for the given data X → $\hat{y} = X\hat{\beta} = \boxed{[X(X^T X)^{-1} X^T]} \underset{\sim}{Y}$

Orthogonal Projection Matrix for $C(X)$ ←

$$= P_X \underset{\sim}{Y}$$

Remarks :- 1.) We cannot observe the true error, i.e., ε

2.) But, we can find the estimated error as :- $\underset{\sim}{e} = \underset{\sim}{y} - \hat{\underset{\sim}{y}} = \underset{\sim}{y} - P_X \underset{\sim}{y}$

$$= (I_n - P_X)\underset{\sim}{y}$$

3.) $P_X = P_X^T$ (Symmetric) 4.) $P_X = P_X^2$ (Idempotent)

5.) $P_X(I_n - P_X) = O$ Matrix 6.) $\underset{\sim}{e}^T \hat{\underset{\sim}{y}} = \underset{\sim}{y}^T (I_n - P_X)^T P_X \underset{\sim}{y} = 0$

Now, we know : $\hat{\beta} = \underbrace{\boxed{(X^T X)^{-1} X^T}}_{\text{Non Stochastic}} Y$ where $Y \sim N(X\beta, \sigma^2 I_n)$

We want to find → $\hat{\underset{\sim}{\beta}} \sim \ ?$

Now, if $Z \sim N(\mu, \Sigma)$, then $AZ \sim N(A\mu, A\Sigma A^T)$

$$\therefore \hat{\beta} \sim N[\cancel{(X^T X)^{-1} X^T X}\beta, \boxed{(X^T X)^{-1} X^T \sigma^2 I_n [(X^T X)^{-1} X^T]^T}]$$

$$= \cancel{(X^T X)^{-1} X^T X}[(X^T X)^{-1}]^T = (X^T X)^{-1} \quad \hookleftarrow$$

So, $\hat{\underset{\sim}{\beta}} \sim N(\underset{\sim}{\beta}, \sigma^2 (X^T X)^{-1})$

Now, $\underset{\sim}{e} = (I_n - P_X)\underset{\sim}{y}$ & $\underset{\sim}{y} \sim N(X\underset{\sim}{\beta}, I_n\sigma^2)$

We want to find $\longrightarrow$ $\underset{\sim}{e} \sim$?

Can be seen that this is X

It can be seen that : $(I_n - P_x) X \underset{\sim}{\beta} = (X - P_x X)\underset{\sim}{\beta} = 0$

Similarily, $(I_n - P_x) I_n \sigma^2 (I_n - P_x)^T = \sigma^2 (I_n - P_x)(I_n^T - P_x^T)$
$= \sigma^2 (I_n - P_x)$

$\therefore \underset{\sim}{e} \sim N(0, (I_n - P_x)\sigma^2)$

Now, $\underset{\sim}{\hat{y}} = X\hat{\underset{\sim}{\beta}} = P_x \underset{\sim}{y}$ [Predictor] & $\underset{\sim}{e} = (I_n - P_x)\underset{\sim}{y}$ [Estimated Error]

$\therefore Cov(\underset{\sim}{\hat{y}}, \underset{\sim}{e}) = Cov(P_x \underset{\sim}{y}, (I_n - P_x)\underset{\sim}{y})$ ☆

We know that $Cov(AX, BX) = A \Sigma B^T$ if $D(AX) = A\Sigma A^T$

So ☆ will give $\longrightarrow$ $P_x \sigma^2 I_n (I_n - P_x)^T = 0$

This means that $\underset{\sim}{\hat{y}}$ & $\underset{\sim}{e}$ are Independently Distributed

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As $E(\hat{\underset{\sim}{\beta}}) = \underset{\sim}{\beta}$; Hence $\hat{\underset{\sim}{\beta}}$ is an Unbiased Estimator of $\underset{\sim}{\beta}$

Estimator :- Function of Known Data only

Let $z_1, z_2 \cdots z_n \overset{iid}{\sim} f_\theta(z)$ $\to$ Parameter [Eg:- μ & σ^2 in case of Normal Distribution]

If an estimator $T(z)$ can be expressed as a linear combination of z with non-random coefficients, i.e., $T(z) = \sum_i \alpha_i z_i$, then $T(z)$ is called a Linear Estimator

Assuming $T(z_1, z_2 \cdots z_n)$ is an Estimator of θ, then if :-

$E(T(z_1, z_2 \cdots z_n)) = \theta \ \forall \theta$ in the parameter space, then $T(z_1, z_2 \cdots z_n)$ is called an Unbiased Estimator of θ.

Simple Linear Regression

Linear Estimator : If $T(\underset{\sim}{y}) = \sum \alpha_i y_i = \alpha^T \underset{\sim}{y}$, then $T(\underset{\sim}{y})$ is a linear estimator

We want to show that $\longrightarrow$ $\hat{\beta}_1$ is a Linear Estimator

We know that $\hat{\beta}_1 = \dfrac{S_{xy}}{S_{xx}} = \dfrac{\sum (y_i - \bar{y})(x_i - \bar{x})}{S_{xx}} = \dfrac{[\sum (x_i - \bar{x}) y_i) - \bar{y}(\sum(x_i - \bar{x}))]}{S_{xx}}$ 0

$\searrow \alpha_i$

$\therefore \hat{\beta}_1 = \sum_{i=1}^{n} \alpha_i y_i \longrightarrow$ Linear Estimator

$\hat{}$ T

$\beta_1 = \alpha' \underset{\sim}{y}$; $\alpha_i = \dfrac{(x_i - \bar{x})}{S_{xx}}$

$\underset{\sim}{\hat{\beta}_1} \sim N(\alpha^T X \beta, \underset{\sim}{\alpha}^T (I_n \sigma^2) \underset{\sim}{\alpha})$; Now $\alpha^T X \beta = \underset{\sim}{\alpha}^T [\underset{\sim}{1} \; \underset{\sim}{x}] \underset{\sim}{\beta} = [\underset{\sim}{\alpha}^T \underset{\sim}{1} \; \underset{\sim}{\alpha}^T \underset{\sim}{x}] \underset{\sim}{\beta}$

Here, $\underset{\sim}{\alpha}^T \underset{\sim}{1} = \sum \alpha_i = 0$ & $\underset{\sim}{\alpha}^T \underset{\sim}{x} = \frac{1}{S_{xx}} \sum (x_i - \bar{x}) x_i = \frac{1}{\cancel{S_{xx}}} \cancel{\sum (x_i - \bar{x})(x_i - \bar{x})} = 1$

$\therefore \alpha^T X \beta = [0 \;\; 1] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \beta_1$

Similarily, $\alpha^T (I_n \sigma^2) \alpha = \sigma^2 \alpha^T \alpha = \sigma^2 \sum \alpha_i^2 = \sigma^2 \frac{1}{S_{xx}^2} \sum (x_i - \bar{x})^2$

$= \dfrac{\sigma^2}{S_{xx}}$

$\therefore \underset{\sim}{\hat{\beta}_1} \sim N\left(\beta_1, \dfrac{\sigma^2}{S_{xx}}\right)$

Note :- 1.) It can be shown that $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

2.) $\hat{\beta}_0$ is a Linear & Unbiased Estimator.

3.) $\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$

Prediction Line

$- \; \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x$

$\therefore \hat{y} = \bar{y} + \hat{\beta}_1 (x - \bar{x})$; Clearly Prediction Line passes through $(\bar{x}, \bar{y})$

Estimation of σ^2

$\underset{\sim}{\varepsilon} \sim N(0, \sigma^2 I_n) \rightarrow$ Not observable

But $\underset{\sim}{e} = (y - \hat{y})^T (y - \hat{y}) = (I_n - P_X) y$

$\sim N(0, (I_n - P_X)\sigma^2)$

Now, $\sum e_i^2 = (y - \hat{y})^T (y - \hat{y}) = (y - \hat{\beta} X)^T (y - \hat{\beta} X) = \sum (y_i - (\beta_0 + \beta_1 x_i))^2$

Although ε's are iid, e_i's are dependent

$\sum e_i^2 = \underset{\sim}{e}^T \underset{\sim}{e} = [(I_n - P_X) \underset{\sim}{y}]^T [(I_n - P_X) \underset{\sim}{y}] = \underset{\sim}{y}^T (I_n - P_X)^T (I_n - P_X) \underset{\sim}{y}$

$= \underset{\sim}{y}^T (I_n - P_X) y^T$

Now, we can write $e^T e = \sigma^2 \left(\frac{e}{\sigma}\right)^T \left(\frac{e}{\sigma}\right) = \sigma^2 \left(\frac{\underset{\sim}{y}}{\sigma}\right)^T (I_n - P_X) \left(\frac{\underset{\sim}{y}}{\sigma}\right)$

By Cochran's Theorem, the above expression will have Chi - Squared Distribution

$$\frac{\underset{\sim}{y}}{\sigma} \sim N\left(\frac{X\beta}{\sigma}, I_n\right); \quad \sigma^{-2}\left(\frac{\underset{\sim}{y}}{\sigma}\right)'(I_n - P_X)\left(\frac{\underset{\sim}{y}}{\sigma}\right) \sim \chi^2$$

$D.o.F = Rank(I_n - P_X)$

$n.c.p = \left(\frac{X\beta}{\sigma}\right)^T (I_n - P_X)\left(\frac{X\beta}{\sigma}\right)$

$$n.c.p = \frac{\beta^T X^T}{\sigma} \times (I_n - P_X)\frac{X\beta}{\sigma} = \frac{\beta^T}{\sigma}(\cancel{X^T X} - \cancel{X^T X})\frac{\beta}{\sigma}$$

$$= 0$$

$$D.o.f. = Rank(I_n - P_X) = n-2$$

$$E(e^T e) = \sigma^2(D.o.f + n.c.p.) = \sigma^2(n-2) \Rightarrow E\left(\frac{e^T e}{n-2}\right) = \sigma^2$$

Let $\hat{\sigma}^2 = \frac{e^T e}{n-2} \Rightarrow E(\hat{\sigma}^2) = \sigma^2$

Thus, $\hat{\sigma}^2$ is an Unbiased Estimator of σ^2.

$$\sum e_i^2 = \sum (y_i - (\beta_0 + \beta_1 x_i))^2 = \sum_i (y_i - (\bar{y} - \hat{\beta}_1\bar{x} - \hat{\beta}_1 x_i))^2$$

$$= \sum_i [(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})]^2 = s_{yy} + \hat{\beta}_1^2 s_{xx} - 2\hat{\beta}_1 s_{xy}$$

$$= s_{yy} + \frac{s_{xy}^2}{s_{xx}^2} \times s_{xx} - 2\frac{s_{xy}}{s_{xx}} s_{xy}$$

$$\therefore \sum e_i^2 = s_{yy} - \frac{s_{xy}^2}{s_{xx}}$$

$H_0: \beta_1 = b_1$ vs $H_1: \beta_1 \neq b_1$; b_1 is given in Qs"

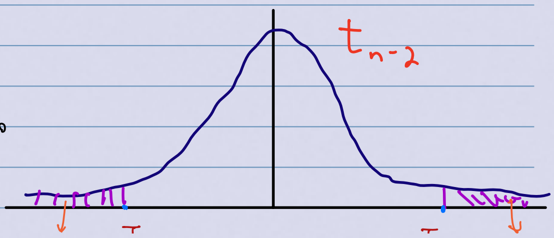
We know that $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right) \Rightarrow \hat{\beta}_1 - \beta_1 \sim N\left(0, \frac{\sigma^2}{S_{xx}}\right)$

$$\therefore \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{S_{xx}}}} \sim N(0,1)$$

If H_0 / Null Hypothesis is true $\longrightarrow \dfrac{\hat{\beta}_1 - b_1}{\sqrt{\frac{\sigma^2}{S_{xx}}}} \sim N(0,1)$ under H_0

We know :- $\hat{\sigma}^2 = \frac{e^T e}{n-2}$

So, $\dfrac{\hat{\beta}_1 - b_1}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}} \sim t_{n-2}$ under H_0



We reject H_0 in favour of H_1 if :-

$|T_{observed}| > \tau_{\frac{\alpha}{2}, n-2}$ at size α $\quad$ $\frac{\alpha}{2}$ $-t_{\frac{\alpha}{2}, n-2}$ $\quad$ $t_{\frac{\alpha}{2}, n-2}$ $\frac{\alpha}{2}$ [$\alpha \rightarrow$ Pre specified size of Test]

SLR

$$Y \sim N(X\beta, \sigma^2 I_n) \; ; \; \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \; \& \; \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Data $\longrightarrow \{(x_i, y_i) \mid i = 1, 2 \cdots n\}$

Now $\underset{\text{Independent}}{y_i \sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$ $\rightarrow$ Not i.d. due to x_i ; so for (x_0, y_0): $\underset{\text{Unknown}}{y_0} \sim N(\beta_0 + \beta_1 \underset{\text{Known}}{x_0}, \sigma^2)$

$\hat{y}_0$ is the Predicted value of y_0 given x_0 is known.
$\hookrightarrow \hat{\beta}_1 x_0 + \hat{\beta}_0$

$E(\hat{y}_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$ [Not y_0 as it itself is a R.V.] $= E(y_0)$

$$Var(\hat{y}_0) = Var(\hat{\beta}_0 + \hat{\beta}_1 x_0) = Var(\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_0) = Var(\bar{y} + \hat{\beta}_1(x_0 - \bar{x}))$$

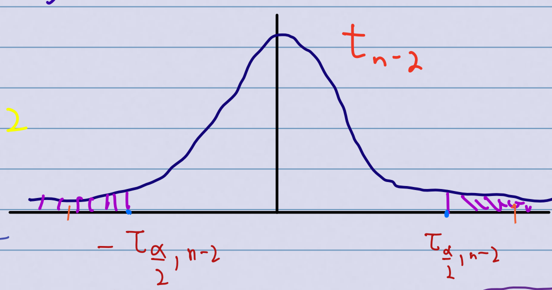
$$= Var\Big(\frac{1}{n} \underset{\sim}{1}^T \underset{\sim}{Y} + \underbrace{(x_0 - \bar{x}) \frac{S_{xy}}{S_{xx}}}_{\alpha^T \hat{Y}}\Big) = \frac{1}{n^2} \times n\sigma^2 + \frac{(x_0 - \bar{x})^2}{S_{xx}^2} \times S_{xx}\sigma^2 + 0 \rightarrow Cov(1^T, \alpha^T) = 0$$

y_0 is a random variable but $E(y_0)$ is a Constant for given x_0

95% confidence interval of $E(y_0) = \beta_0 + \beta_1 x_0$

$$\hat{y}_0 \sim N\Big(\beta_0 + \beta_1 x_0, \sigma^2\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)\Big)$$

$$\Rightarrow \frac{\hat{y}_0 - (\beta_0 + \beta_1 x_0)}{\sqrt{\sigma^2\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)}} \propto t_{n-2}$$



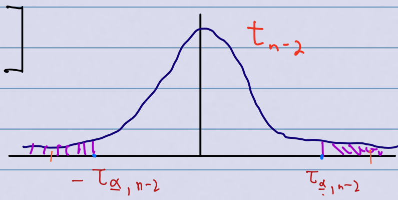
95% Confidence Interval of $(\beta_0 + \beta_1 x_0)$:-

$$\Big(\hat{\beta}_0 + \hat{\beta}_1 x_0 - \tau_{\alpha/2, n-2}\sqrt{\sigma^2\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)}, \hat{\beta}_0 + \hat{\beta}_1 x_0 + \tau_{\alpha/2, n-2}\sqrt{\sigma^2\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)}\Big)$$

Let y_0 & $(y_1, y_2 \cdots y_n)$ be Independent.

$$\Rightarrow y_0 - \hat{y}_0 \sim N\Big[0, \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)\Big]$$

$$\therefore T_0 = \frac{y_0 - \hat{y}_0}{\sqrt{\hat{\sigma}^2\Big(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big)}} \sim t_{n-2}$$



$$P\left(-\tau_{\alpha/2, n-2} < T_0 < \tau_{\alpha/2, n-2}\right) = 1-\alpha = 0.95$$

$\therefore$ 95 % Prediction Interval of y_0 is :- $\left[\hat{y}_0 \mp \tau_{\alpha/2, n-2}\sqrt{\hat{\sigma}^2\left(1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}\right]$

When Regressor is Stochastic

We know that :- $$f(x,y) = \frac{\exp\left[-\frac{1}{2}\frac{1}{(1-\rho^2)} \times Q(x,y)\right]}{2\pi\,\sigma_x\sigma_y\sqrt{1-\rho^2}}$$

Here $$Q(x,y) = \left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)$$

Rearranging the terms $\longrightarrow$

$$Q(x,y) = \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right) + \rho^2\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + (1-\rho^2)\left(\frac{x-\mu_x}{\sigma_x}\right)^2$$

$$= \left[\left(\frac{y-\mu_y}{\sigma_y}\right) - \rho\left(\frac{x-\mu_x}{\sigma_x}\right)\right]^2 + (1-\rho^2)\left(\frac{x-\mu_x}{\sigma_x}\right)^2$$

$$\therefore Q(x,y) = \frac{1}{\sigma_y^2}\left[y-\mu_y-\rho\frac{\sigma_y}{\sigma_x}(x-\mu_x)\right]^2 + (1-\rho^2)\left(\frac{x-\mu_x}{\sigma_x}\right)^2$$

Substituting $Q(x,y)$ in $f(x,y)$ we get :-

$$f(x,y) = \frac{\exp\left[-\frac{1}{2}\frac{1}{(1-\rho^2)} \times \left(\frac{1}{\sigma_y^2}\left[y-\mu_y-\rho\frac{\sigma_y}{\sigma_x}(x-\mu_x)\right]^2 + (1-\rho^2)\left(\frac{x-\mu_x}{\sigma_x}\right)^2\right)\right]}{\sqrt{2\pi}\left(\sigma_y\sqrt{1-\rho^2}\right)\sqrt{2\pi}\,\sigma_x}$$

$$= \underbrace{\frac{e^{-\frac{1}{2}\frac{1}{(1-\rho^2)} \times \frac{1}{\sigma_y^2}\left[y-\left(\mu_y+\rho\frac{\sigma_y}{\sigma_x}(x-\mu_x)\right)\right]^2}}{\sqrt{2\pi}\sqrt{\sigma_y^2(1-\rho^2)}}}_{\text{Conditional}} \times \underbrace{\frac{e^{-\frac{1}{2}\left(\frac{x-\mu_x}{\sigma_x}\right)^2}}{\sqrt{2\pi}\,\sigma_x}}_{\text{Marginal}}$$

Hence $Y|X \sim N\left(\mu_y+\rho\frac{\sigma_y}{\sigma_x}(x-\mu_x),\ \sigma_y^2(1-\rho^2)\right)$

$Var(Y) = \sigma_y^2 \geq (1-\rho^2)\sigma_y^2 = Var(Y|X)$

$$\begin{pmatrix}\underset{\sim}{Y}\\ \underset{\sim}{X}\end{pmatrix} \sim N\left[\begin{pmatrix}\underset{\sim}{\mu_y}\\ \underset{\sim}{\mu_x}\end{pmatrix}, \begin{pmatrix}\Sigma_{yy} & \Sigma_{yx}\\ \Sigma_{yx}^T & \Sigma_{xx}\end{pmatrix}\right]$$; $Y|X_{=x} \sim N\left(\underset{\sim}{\mu_y}+\Sigma_{yx}^T\Sigma_{xx}(X-\underset{\sim}{\mu_x}),\ \Sigma_{yy}-\Sigma_{yx}^T\Sigma_{xx}^{-1}\Sigma_{yx}\right)$

Now, If $(X,Y) \sim BVN(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho)$ [To check Independence of X & Y]

① $H_0 : \rho = 0$ (Independent)

v.s

For Testing

$H_1 : \rho \neq 0$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

(Sample correlation)

$\Downarrow$

$$T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \quad \text{Under } H_0$$

② $H_0 : \rho = \rho_0 \neq 0$

$H_1 : \rho \neq \rho_0$

$$Z = \ln\left(\frac{1+r}{1-r}\right) \quad \tanh^{-1} \text{ Transformation}$$

$$\mu_0 = \ln\left(\frac{1+\rho_0}{1-\rho_0}\right) \quad \& \quad \sigma_Z^2 = (n-3)^{-1}$$

$$\boxed{W = \frac{Z - \mu_0}{\sigma_Z} \sim N(0,1) \text{ as } n \longrightarrow \infty}$$

Asymptotic Test

Multiple Linear Regression

$\underset{\sim}{Y} \sim N(X\underset{\sim}{\beta}, \sigma^2 I_n)$; $\underset{\sim}{Y} \in \mathbb{R}^n$, $\underset{\sim}{\varepsilon} \in \mathbb{R}^n$

$\underset{n\times(k+1)}{X} \in \mathbb{R}^{n\times(k+1)}$, $\underset{\sim}{\beta} \in \mathbb{R}^{k+1}$, $\sigma^2 > 0$

Here, there are $k+2$ unknown parameters, $\underset{\sim}{\beta}$ & $\sigma^2 > 0$

☆ Similar to SLR, after we Minimise $\beta_{L.S} = \hat{\beta}$

$\hat{\beta} = (X^T X)^{-1} X^T Y$; So $\hat{y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y$

$= P_X Y \longrightarrow$ Projection of y in $C(X)$

$\underset{\sim}{e} = y - \hat{y} = (I_n - P_X)\underset{\sim}{Y}$; $\underset{\sim}{e}^T \hat{\underset{\sim}{Y}} = 0$

$\hat{y} \sim N(X\beta, \sigma^2 P_X)$; $\underset{\sim}{e} \sim N(0, \sigma^2(I_n - P_X))$

Here $e \in C(X)^\perp = C(X^T X)^\perp$

Here, $\hat{y}$ & e are Uncorrelated & they are independently distributed when $\varepsilon \sim N(0, \sigma^2 I_n)$

Linear Unbiased Estimator

A linear estimator $l^T y = \sum l_i y_i$ is said to be a **LUE** of $p^T \beta$ if $E(l^T y) = p^T \beta$ for all $\beta \in \mathbb{R}^{k+1}$

Linear Zero Function

A linear estimator $l^T y = \sum l_i y_i$ is said to be a LZF if $E(l^T y) = 0$ for all $\beta \in \mathbb{R}^{k+1}$

BLUE

The Best Linear Unbiased Estimator (BLUE) of a linear parameter function $p^T \beta$ is a LUE with **minimum variance**

Theorem - A linear function is BLUE of its expectation iff it is uncorrelated with all linear zero function (LZF)

☆ If $l^T y$ is an LUE of $p^T \beta$, then $l^T P_x y$ is the BLUE of $p^T \beta$

Analysis of Variance [ANOVA]

$H_0 : \beta_1 = \beta_2 \cdots = \beta_k = 0$

$H_1 :$ H_0 is not true

we use $\bar{y}$ to estimate it.

→ Under H_0, $y_i \overset{iid}{\sim} N(\beta_0, \sigma^2)$

$$SS_{Total} = \sum (y_i - \bar{y})^2 = \underset{\sim}{Y}^T \left(I_n - \frac{1}{n} \underset{\sim}{1}\,\underset{\sim}{1}^T\right) \underset{\sim}{Y}$$ [Model building is not needed]

$$SS_{Error} = \sum (y_i - \hat{y}_i) = (y - \hat{y})^T (y - \hat{y}) = [(I_n - P_x)\underset{\sim}{Y}]^T (I_n - P_x)\underset{\sim}{Y}$$

$$= \underset{\sim}{Y}^T (I_n - P_x) \underset{\sim}{Y}$$

$$SS_{Model} = \sum (\hat{y}_i - \bar{y})^2 = \underset{\sim}{Y}^T \left(P_x - \frac{1}{n} \underset{\sim}{1}\,\underset{\sim}{1}^T\right) \underset{\sim}{Y}$$

We can notice that :- $SS_{Total} = SS_{Error} + SS_{Model}$

By Cochran's Theorem :- $\dfrac{SS_{Error}}{\sigma^2} \sim \chi^2_{d.f. = n-k-1}$ [Rank $(I_n - P_x)$]

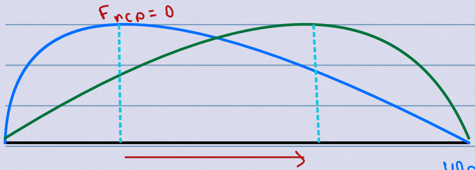
ncp = 0

$\dfrac{SS_{Model}}{\sigma^2} \sim \chi^2_{d.f. = k}$ $\qquad$ $\dfrac{SS_{Total}}{\sigma^2} \sim \chi^2_{d.f. = n-1}$

$ncp = \lambda \geq 0$ $\qquad\qquad$ $ncp = \lambda \geq 0$

$$\left.\begin{array}{ll} \text{Under } H_0: & \lambda = 0 \\ \phantom{\text{Under }} H_1: & \lambda > 0 \end{array}\right] \Rightarrow \frac{\frac{SS_{Model}}{\sigma^2} / k}{\frac{SS_{Error}}{\sigma^2} / (n-k-1)} \sim F_{k, n-k-1 ;\, ncp = 0} \quad \text{Under } H_0$$

$$F_{k, n-k-1 ;\, ncp = \lambda > 0} \quad \text{Under } H_1$$



We will reject H_0 In favour of H_1 if :-

$$F_{obs} > F_{\alpha, k, n-k-1}$$

★ $H_0 : \beta_j = b_j$; $\hat{\beta}_j = \boxed{\underset{\sim}{\ell}_j^T} \hat{\underset{\sim}{\beta}} \sim N(\ell_j^T \beta, \sigma^2 (\ell_j))$

$H_1 : \beta_j \neq b_j$ $\qquad = N(\beta_j, \sigma^2 (C_{jj}))$

$C = (X^T X)^{-1}$

$$\underset{\sim}{\ell}_j = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \rightarrow j^{th}$$

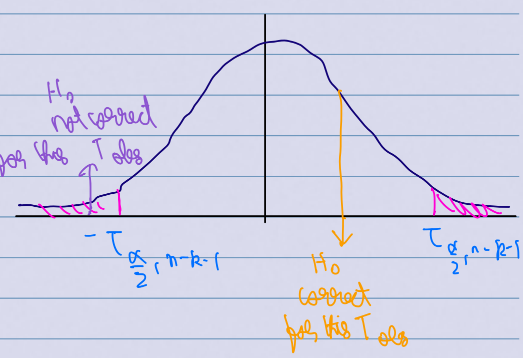
Thus, $\hat{\beta}_j - \beta_j \sim N(0, \sigma^2 C_{jj}) \Rightarrow \dfrac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2 C_{jj}}} \sim N(0,1)$

But σ is Unknown.

$$\hat{\sigma}^2 = \frac{SS_{Error}}{n-k-1} = MS_{Error} = \frac{\underset{\sim}{Y}^T (I_n - P_X) Y}{n-k-1}$$

$$\therefore \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} = \frac{(\hat{\beta}_j - \beta_j) / \sqrt{\sigma^2 C_{jj}}}{\sqrt{\left(\frac{\underset{\sim}{Y}^T (I_n - P_X) Y}{\sigma^2} \right) / (n-k-1)}} \sim t_{n-k-1}$$

Under H_0 : $\dfrac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t_{n-k-1}$



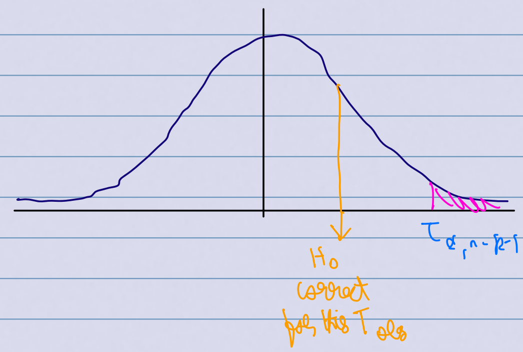
★★ $H_0 : \beta_i - \theta \beta_j = 0$; $\underset{\sim}{\ell} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ -\theta \\ \vdots \\ 0 \end{pmatrix} \begin{array}{l} \\ \\ \\ \rightarrow i \\ \\ \rightarrow j \\ \\ \\ \end{array}$

$H_1 : \beta_i - \theta \beta_j \neq 0$

$$\beta_i - \theta \beta_j = \underset{\sim}{\ell}^T \underset{\sim}{\beta}$$

$$\underset{\sim}{l}^T \hat{\underset{\sim}{\beta}} \sim N(\underset{\sim}{l}^T \underset{\sim}{\beta}, \sigma^2 \underset{\sim}{l}^T C \underset{\sim}{l}) = N(\beta_i - \theta\beta_j, \sigma^2(C_{ii} - 2\theta C_{ij} + \theta^2 C_{jj}))$$

So,
$$\frac{\underset{\sim}{l}^T \hat{\underset{\sim}{\beta}} - \underbrace{\underset{\sim}{l}^T \underset{\sim}{\beta}}_{b}}{\sqrt{\hat{\sigma}^2 (C_{ii} - 2C_{ij}\theta + \theta^2 C_{jj})}} \sim t_{n-k-1} \quad \text{Under } H_0$$



We reject H_0 in favour of H_1 at level α iff :-

$$T_{obs} > T_{\alpha, n-k-1}$$

★★★ $H_0 : \begin{cases} \beta_i = 0 \\ \beta_j = 0 \end{cases}$ or $H_0 : \begin{cases} \beta_i = 0 \\ \beta_i - \beta_j = 0 \end{cases}$

$$A \underset{\sim}{\beta} = 0 ; \quad A = \begin{array}{c} \quad\quad\quad i \\ \begin{pmatrix} 0 & 0 & \cdots & 1 & \cdots & & 0 \\ 0 & 0 & \cdots & 1 & \cdots & 0 & 0 \; 0 \end{pmatrix} \\ \quad\quad\quad j \end{array}$$

$$B \underset{\sim}{\beta} = 0 ; \quad B = \begin{array}{c} \quad\quad i \\ \begin{pmatrix} 0 & 0 & \cdots & 1 & \cdots & \cdots & 0 \\ 0 & 0 & \cdots & 1 & \cdots & -1 & \cdots & 0 \end{pmatrix} \\ \quad\quad i \quad\quad j \end{array}$$

$$A \hat{\underset{\sim}{\beta}} \sim N(A \underset{\sim}{\beta}, \sigma^2 A C A^T) \Rightarrow A\hat{\underset{\sim}{\beta}} - A\underset{\sim}{\beta} \sim N(0, \sigma^2 A C A^T)$$

By Cochran's Theorem :- $(A\hat{\underset{\sim}{\beta}} - A\underset{\sim}{\beta})^T (\sigma^2 A C A^T)^{-1} (A\hat{\underset{\sim}{\beta}} - A\underset{\sim}{\beta}) \sim \chi^2_{Rank(A)}$; $ncp = 0$

$$\therefore \frac{(A\hat{\underset{\sim}{\beta}} - A\underset{\sim}{\beta})^T (A C A^T)^{-1} (A\hat{\underset{\sim}{\beta}} - A\underset{\sim}{\beta}) / Rank(A)}{(Y^T(I_n - P_X)Y) / (n-k-1)} \sim F_{\substack{df_1 = Rank(A) \\ df_2 = n-k-1}}$$

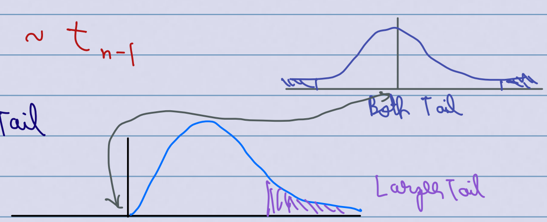
Under H_0 :
$$\frac{(A\hat{\underset{\sim}{\beta}} - b)^T (A C A^T)^{-1} (A\hat{\underset{\sim}{\beta}} - b) / Rank(A)}{(Y^T(I_n - P_X)Y) / (n-k-1)} \sim F_{\substack{df_1 = Rank(A) \\ df_2 = n-k-1}}$$

$b = 0$ in our case ; Now for testing : $X \sim N(\mu, \sigma^2)$

$H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$

Note :- $T = \dfrac{\sqrt{n}(\bar{X} - \mu_0)}{\hat{\sigma}} \sim t_{n-1}$

$T^2 \sim F_{1, n-1} \Rightarrow$ Only upper Tail



★★★ $H_0 : \beta_1 = \beta_2 = \cdots \beta_k = 0$
H_1 : H_0 is not true.

$$A = \begin{bmatrix} 0 & \uparrow & \\ 0 & & k \\ 0 & & \\ \vdots & & \\ 0 & \downarrow & \end{bmatrix}_{k\times(k+1)} \quad X \ \underset{\sim}{\beta} = \underset{\sim}{0}$$

Under H_0 :

$$\frac{(A\hat{\underset{\sim}{\beta}} - 0)^T (A C A^T)^{-1} (A\hat{\underset{\sim}{\beta}} - 0) / \xrightarrow{k} \cancel{\text{Rank}}(A)}{(Y^T(I_n - P_X)Y)/(n-k-1)} \sim F_{df_1 = k,\ df_2 = n-k-1}$$

We can also write the F statistic in another way.

$$F = \frac{\frac{SS_{Model}}{\sigma^2} / k}{\frac{SS_{Error}}{\sigma^2} / (n-k-1)} \quad ; \quad \underset{\substack{(n-1)\\ ncp(\lambda)}}{SS_{Total}} = \underset{\substack{(k)\\ ncp(\lambda)}}{SS_{Model}} + \underset{\substack{(n-k-1)\\ ncp(0)}}{SS_{Error}}$$

Now, we know : $\frac{\underset{\sim}{Y}}{\sigma} \sim N\left(\frac{X\beta}{\sigma}, I_n\right)$

$$SS_{Model} = \sum (\hat{y}_i - \bar{y})^2 = \underset{\sim}{Y}^T \left(P_X - \frac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\right)\underset{\sim}{Y}$$

So, $ncp\left(\frac{SS_{Model}}{\sigma^2}\right) = \left(\frac{X\beta}{\sigma}\right)^T \left(P_X - \frac{1}{n}\underset{\sim}{1}\underset{\sim}{1}^T\right)\left(\frac{X\beta}{\sigma}\right) = \frac{1}{\sigma^2}\begin{pmatrix}\beta_0\\ \beta_R\end{pmatrix}^T \begin{bmatrix} 0 & 0^T \\ 0 & X_R^T X_R - \frac{1}{n}X_R^T \underset{\sim}{1}\underset{\sim}{1}^T X_R \end{bmatrix}\begin{pmatrix}\beta_0\\ \beta_R\end{pmatrix}$

Here, $\beta = \begin{pmatrix}\beta_0 \\ \beta_R\end{pmatrix}$ & $X = \begin{bmatrix} 1 & \\ \vdots & X_R \\ 1 & \end{bmatrix}$

Now, $\frac{1}{(n-1)}\left[X_R^T X_R - \frac{1}{n} X_R^T \underset{\sim}{1}\underset{\sim}{1}^T X_R\right] = Cov(\underset{\sim}{X})$

For $ncp = 0 \Rightarrow \beta_R = 0$

Polynomial Linear Regression

Consider a linear model $\longrightarrow y_i = \beta_0 + \beta_1 x_i^1 + \beta_2 x_i^2 + \dots \beta_k x_i^k + \epsilon_i \ \forall i = 1, 2 \dots n$
We want to estimate β which will minimise least squared condition.

In Polynomial Linear Regression, we consider Higher degree of component x, but it is linear in parameters.

$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$ For single regressor

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \varepsilon$ For Multiple Regressor

Orthogonal Polynomial

- For a given set of input data $x = (x_1, x_2 \cdots x_n)$, a set of polynomials $\{P_0, P_1, \cdots, P_k\}$ are said to be Orthogonal Polynomial if :-

$$P_0(x_i) = 1 \quad \& \quad \sum P_j(x_i) P_k(x_i) = 0 \quad \forall j \neq k$$

So, our model is $\longrightarrow$ $y_i = \sum_{j=0}^{k} \alpha_j P_j(x_i) + \varepsilon_i \quad \forall i = 1, 2, \cdots n$

$\Rightarrow Y = Z\alpha + \varepsilon$; When $Y \sim N(Z\alpha, \sigma^2 I_n)$, the least square estimate will be :-

$$\hat{\alpha} = (Z^T Z)^{-1} Z^T y$$

Coefficient of Determination

For a given set of data (y, x), we first fit linear model under standard assumptions. To measure how good the model is to explain dependencies between regression & independent variables, we use Coefficient of Determination (R^2)

$$R^2 = \frac{\text{Variation of } Y \text{ explained by the model}}{\text{Total variation in } Y} = \frac{\sum_{i=1}^{n} (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^{n} (Y_i - \bar{Y})^2} = \frac{SS_{Model}}{SS_{Total}}$$

$$= 1 - \frac{SS_{Error}}{SS_{Total}} = 1 - \frac{\sum e_i^2}{\sum (Y_i - \bar{Y})^2}$$

1. $R^2 \in (0, 1)$
2. R^2 is an increasing function of the number of variables
3. Higher value of R^2 implies a better model.

Adjusted R^2 :

$$R^2_{adj} = 1 - \frac{\sum e_i^2 / (n-k-1)}{\sum (Y_i - \bar{Y})^2 / (n-1)} < R^2$$

gives us an appropriate time to stop.